## **Lecture 7** (*March 3*)

*Reading: Section 3.2*
*Viewing: Lectures 6-7*

Consider the subspace $S \subset \mathbb{R}^3$ spanned by the vectors:

$$\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \qquad \text{and} \qquad \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix} \tag{58}$$

These two vectors are not on the same line (passing through the origin) so we conclude that $S$ is a plane. As such, $S$ is cut out by a single equation in the plane, and it is not hard to guess what this equation is:

$$S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ such that } x + y + z = 0 \right\} \tag{59}$$

Indeed, both vectors in (58) have the property that the sum of their entries is 0. Moreover, any linear combination of those two vectors has the property that the sum of its entries is 0, and this allows us to conclude (59). A more linear algebra way to say this is to consider the $1 \times 3$ matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

(the entries of $A$ are precisely the coefficients of $x$, $y$ and $z$ in the equation of the plane (59)) and to observe the fact that:

$$A \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix} \qquad \text{and} \qquad A \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

Moreover, taking appropriate linear combinations of these two equations allows us to conclude that:

$$A \left( \alpha \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + \beta \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix} \right) = \begin{bmatrix} 0 \end{bmatrix}$$

for any scalars $\alpha$ and $\beta$. We can summarize the formula above by saying that the plane $S$ is precisely what people call the nullspace of $A$, as in the following definition.

**Definition 6.** *The* ***nullspace*** *of an $m \times n$ matrix $A$ is the subspace:*

$$N(A) \subset \mathbb{R}^n \tag{60}$$

*consisting of those vectors $\boldsymbol{v} \in \mathbb{R}^n$ such that $A\boldsymbol{v} = 0$.*

**Remark.** *A little quirk concerning our notation: the right-hand side of the equation $A\boldsymbol{v} = 0$ should be an $m \times 1$ column vector, specifically the zero vector $\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$. People usually abbreviate the zero vector by using the notation "$0$", although it is still technically a vector.*

The fact that the nullspace of the matrix $A$ satisfies the requirements (52) and (53) of being a subspace follows from the easy to prove relations:

$$A\boldsymbol{v} = 0 \text{ and } A\boldsymbol{w} = 0 \;\Rightarrow\; A(\boldsymbol{v}+\boldsymbol{w}) = 0$$

$$A\boldsymbol{v} = 0 \text{ and } \alpha \in \mathbb{R} \quad \Rightarrow \quad A(\alpha\boldsymbol{v}) = 0$$

So computing the nullspace of a matrix $A$ boils down to solving the equation $A\boldsymbol{v} = 0$. However, many matrices have the same nullspace. For example, if:

$$\boxed{A = LR} \tag{61}$$

where $L$ is an <u>invertible</u> matrix (such as the product of elimination/permutation/diagonal matrices that you encounter in Gaussian and Gauss-Jordan elimination, hint-hint) then:

$$\boxed{N(A) = N(R)} \tag{62}$$

which is essentially saying that:

$$A\boldsymbol{v} = LR\boldsymbol{v} = 0 \qquad \text{is equivalent to} \qquad R\boldsymbol{v} = 0$$

(the equivalence is showed by multiplying on the left with $L^{-1}$). Gauss-Jordan elimination gives us a recipe for writing $A$ in the form (61), where $R$ is in reduced row echelon form, i.e.:

- the pivots are all 1
- the pivots go to the right as we read the rows from top to bottom
- all the entries above a pivot are zeroes

For example, the following matrix is in reduced row echelon form:

$$R = \begin{bmatrix} \boxed{1} & 3 & 0 & 7 \\ 0 & 0 & \boxed{1} & 2 \end{bmatrix}$$

(as usual, the pivots are in boxes). A column which has a pivot is called a (unsurprisingly) **pivot column** and all other columns are called **free columns**. In the example of $R$ above, the first and third are pivot columns, and the second and fourth are free columns. Explicitly, we have:

$$R\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = 0 \qquad \text{means that} \qquad \begin{bmatrix} a+3b+7d \\ c+2d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \text{i.e.} \qquad \begin{cases} a = -3b-7d \\ c = -2d \end{cases}$$

So here's the catch: for any choice of the numbers $b$ and $d$, the equalities above tell you what $a$ and $c$ have to be in order for the vector:

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

to lie in the nullspace of $R$. Thus, $b$ and $d$ are **free variables**, and they precisely correspond to the free columns of $R$. We conclude that the nullspace of $R$ can be explicitly described as:

$$N(R) = \left\{ \begin{bmatrix} -3b - 7d \\ b \\ -2d \\ d \end{bmatrix} \text{ for all } b, d \text{ real numbers} \right\}$$

The set above is simply a plane in 4-dimensional space. If you want to describe this plane as being spanned by two vectors, then all you need to do is to assign concrete values to the free variables. A reasonable choice is to set $(b, d) = (1, 0)$ or $(0, 1)$, and this implies that:

$$N(R) \text{ is spanned by the vectors } \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} -7 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Therefore, in general, here is the algorithm for computing the nullspace of any $m \times n$ matrix $A$ (or equivalently, for finding all solutions to the system of equations $A\boldsymbol{v} = 0$):

- Write $A$ in reduced row echelon form, and call the resulting matrix $R$

- Identify the pivot columns and the free columns of $R$; the pivot/free variables will be those $x_i$ among $x_1, \ldots, x_n$ such that the $i$–th column of $R$ is a pivot/free column

- The nullspace of $A$, which coincides with the nullspace of $R$, is spanned by those vectors $(x_1, \ldots, x_n)$ where a given free variable is 1, all other free variables are 0, and all the pivot variables are determined by the free variables and the equations:

  $$R \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = 0$$

  It is easy to solve for the pivot variables (in terms of the free variables) in the equation above by back-substitution, precisely because $R$ is in reduced row echelon form. The aforementioned ease is one of the main reasons why people bother with reduced row echelon form in the first place.